Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Hounds of Singularity
b. Team leader name: Harshwardhan Tripathi
c. Problem Statement: Media, Content & Digital Experiences

Team member: Chennika

Brief about the idea

ProofMesh separates the identity of an image from the claims made about it.

The Chrome extension checks visible X posts against a shared investigation registry. A familiar image with a different caption stays unverified.

On demand, Gemini researches the exact claim, assesses fetched source passages and critiques its conclusion. Media origin remains unknown.

Opportunities

Difference: completed investigations become reusable context in the browsing feed, rather than a one-off chat response.

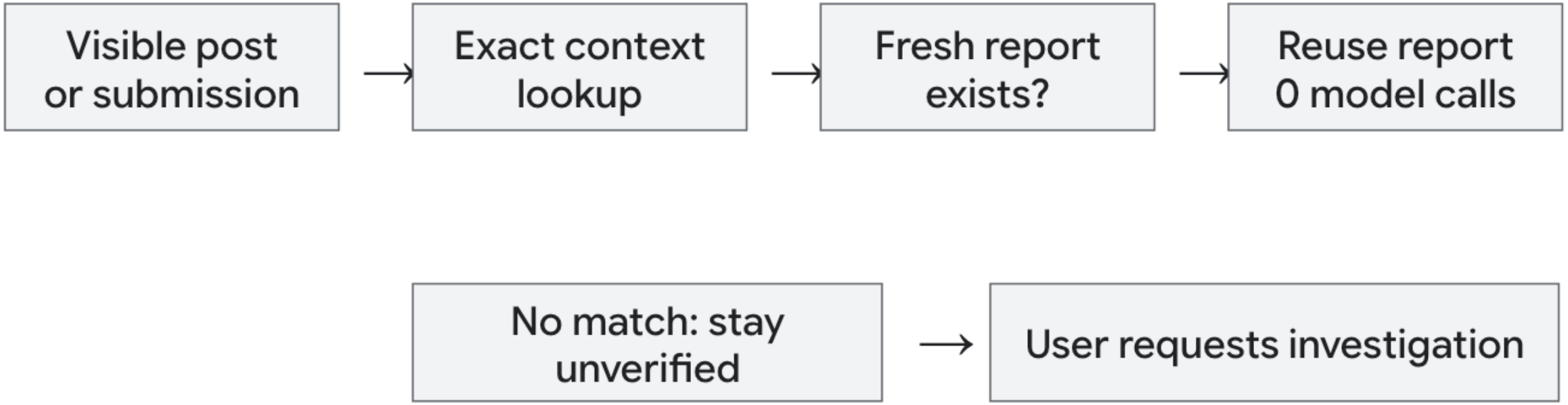
Problem addressed: authentic media can be assigned a misleading date, location or event. Each caption needs its own evidence.

USP: exact-context reuse makes zero model calls. Similar images retrieve leads, never an inherited verdict.

List of features offered by the solution

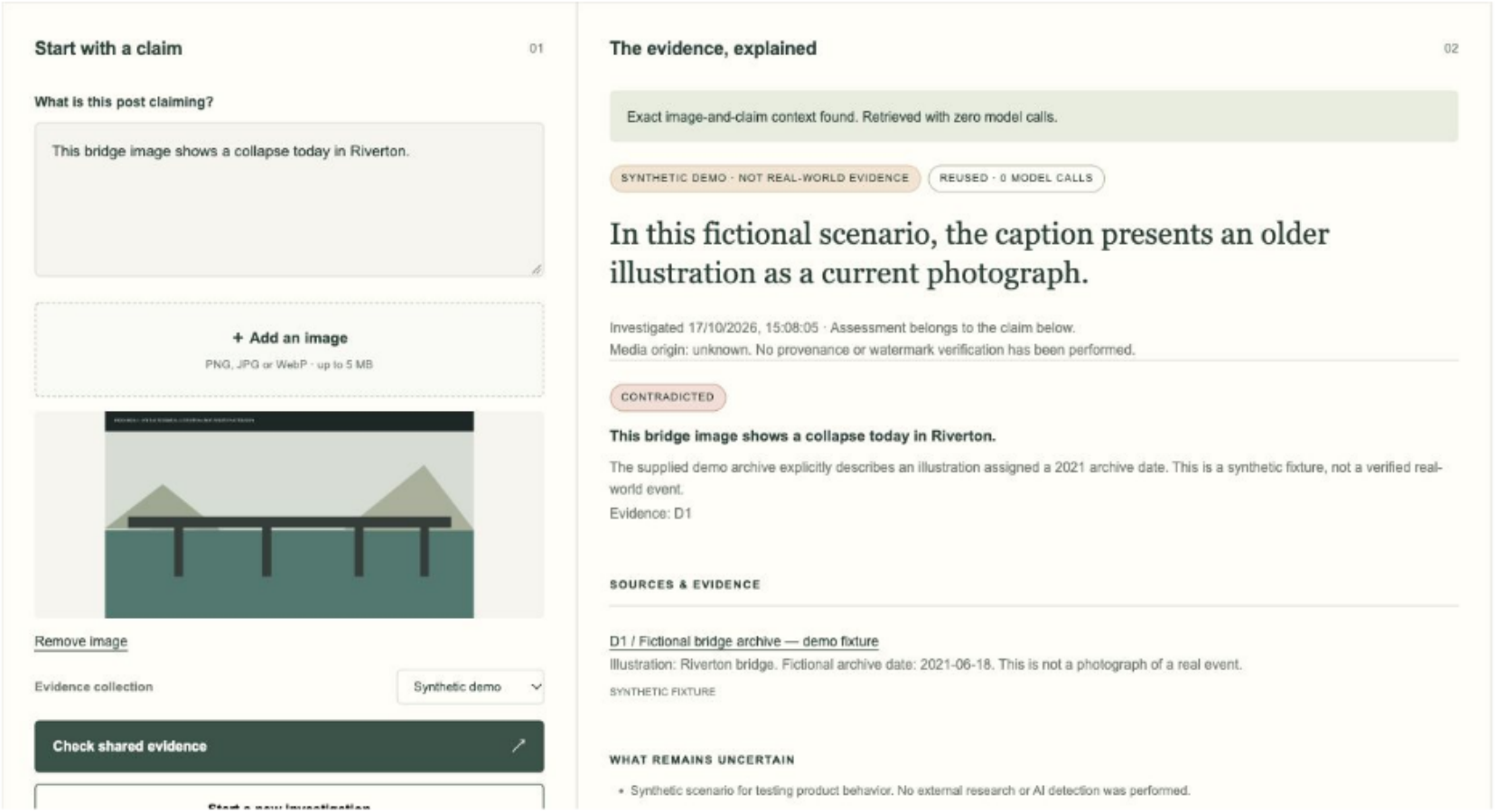
- Visible-post extraction and opt-in registry lookups on X
- Image fingerprints and separate caption identity
- Source-linked supported, contradicted or unresolved findings
- Persistent jobs with polling, deduplication and bounded retry
- Public recorded cases and downloadable evidence reports

Process flow diagram or Use-case diagram

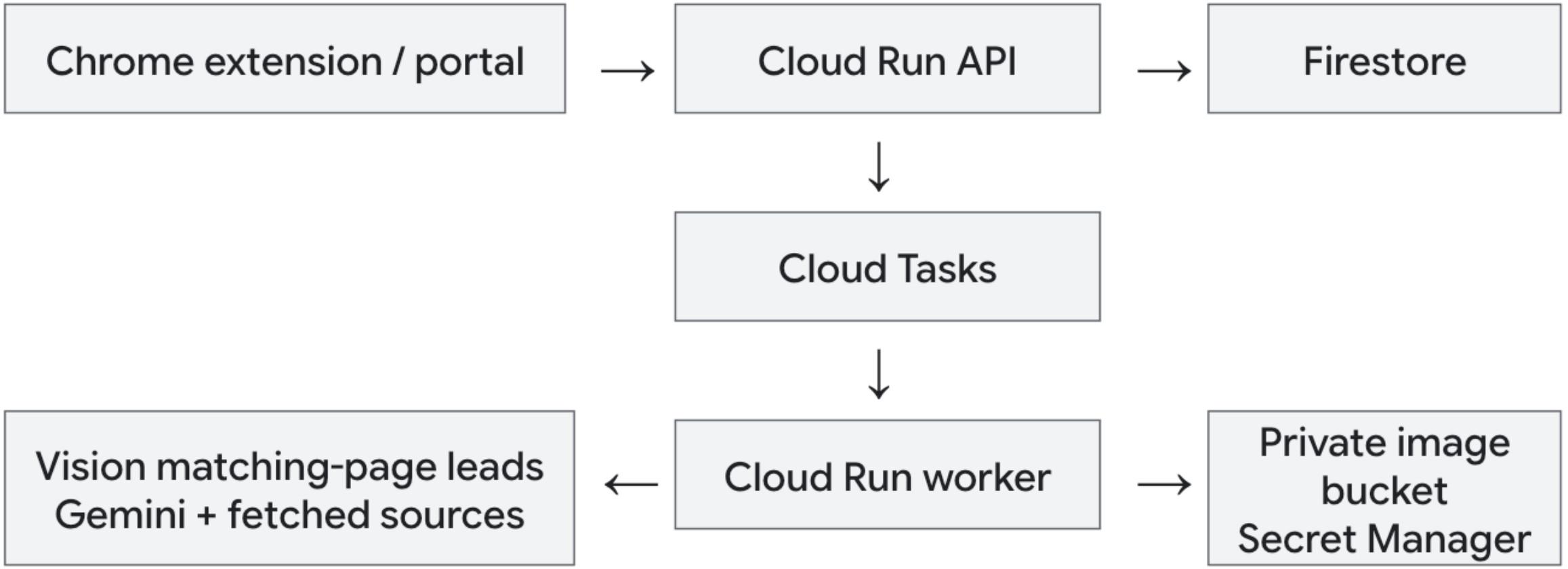


A matching image does not answer a new caption.

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Mumbai resources; Gemini global endpoint. Attached service identities.

Technologies to be used in the solution

Gemini on Vertex AI: search research, structured assessment and critique
Cloud Vision Web Detection: image matching-page leads
Cloud Firestore: reports, jobs and transactional admission limits
Cloud Run + Cloud Tasks: application and durable work delivery
Cloud Storage + Secret Manager: temporary images and credentials
FastAPI, Python, Chrome Manifest V3, HTML/CSS/JavaScript

Estimated implementation cost (optional)

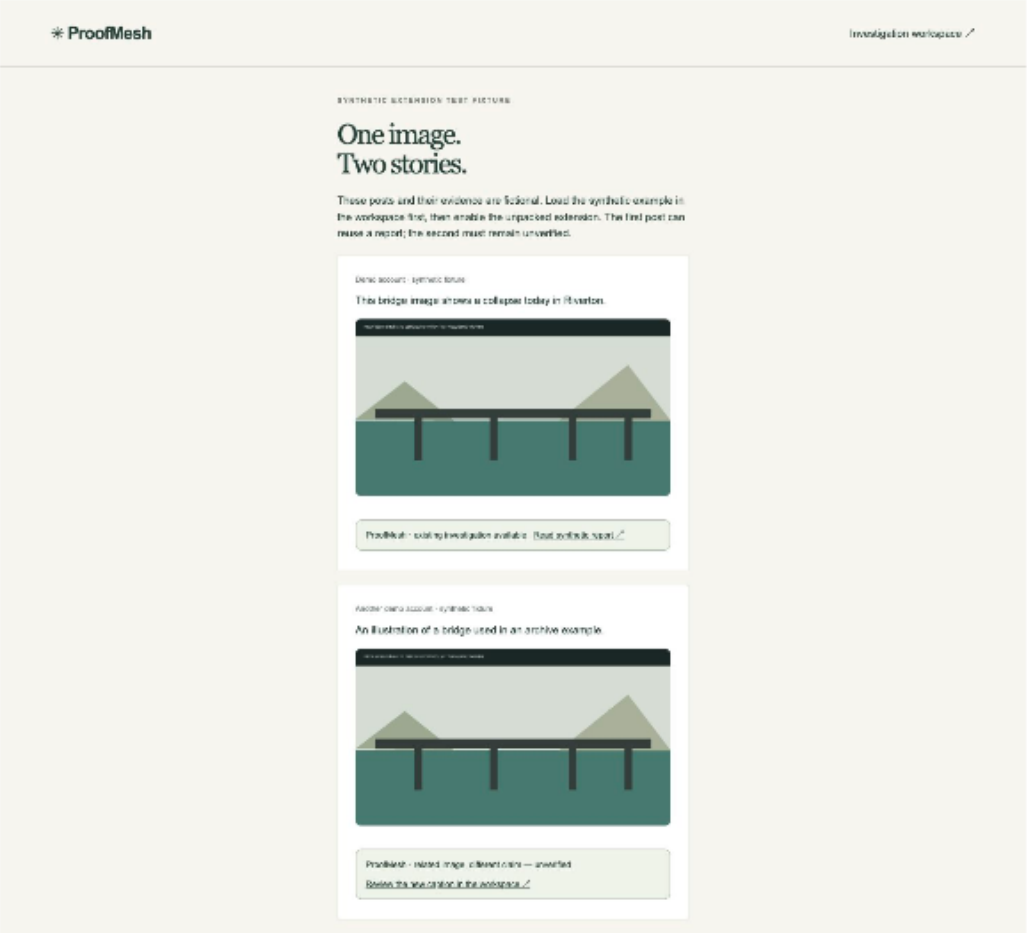
Approved trial-credit budget: US$5 total for bounded work

Four selected completed local runs: estimated model-token cost US$0.0711, before search, Vision, infrastructure and failed attempts. This is not a billing receipt.

Controls: zero minimum / one maximum instance; four new attempts per day and twelve across the live window.

No billing upgrade. Alerts and workload limits are not an absolute dollar cap.

Snapshots of the prototype





Prototype Performance report/Benchmarking

Selected live cases: supported historical caption; contradicted caption on the same image; unresolved assertion with no evidence; source-injection attempt.

Four exact repeats: zero additional model calls.
Local fresh runs: 3.3–25.9 s; local reuse: 1.0–21.0 ms.
Hosted queued run: 10.531 s; Firestore reuse: 104.5 ms.

36 Python + 23 JavaScript tests. Provider failures and one incidental temporal wording error are documented. No accuracy percentage is claimed.

Additional Details/Future Development (if any)

Current limits: X and single-image posts only. No watermark detector, C2PA validation, video analysis or semantic claim reuse.

Same-model critique can miss mistakes; report links and explicit uncertainty remain essential.

Next work: larger labeled evaluation with independent reviewers, broader logged-in X coverage, fingerprint-only lookup and an Instagram-specific adapter.

Project links

Public repository
https://github.com/htg2704/proofmesh

Demo video (3 minutes 20 seconds)
https://proofmesh-2god3tua3a-el.a.run.app/assets/demo.mp4

Working prototype
https://proofmesh-2god3tua3a-el.a.run.app/

Evidence and evaluation notes

Primary historical reference
science.nasa.gov/resource/apollo-8s-iconic-earthrise/
Image credit: NASA/Bill Anders; educational example, no endorsement.

AI assessments are ProofMesh outputs. The misleading-caption recorded case includes a QA note identifying an incidental mission-date wording error.

Full methodology and limits: docs/EVALUATION.md in the repository.
Public cases are recorded runs; viewing them makes no inference calls.

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Thank you

